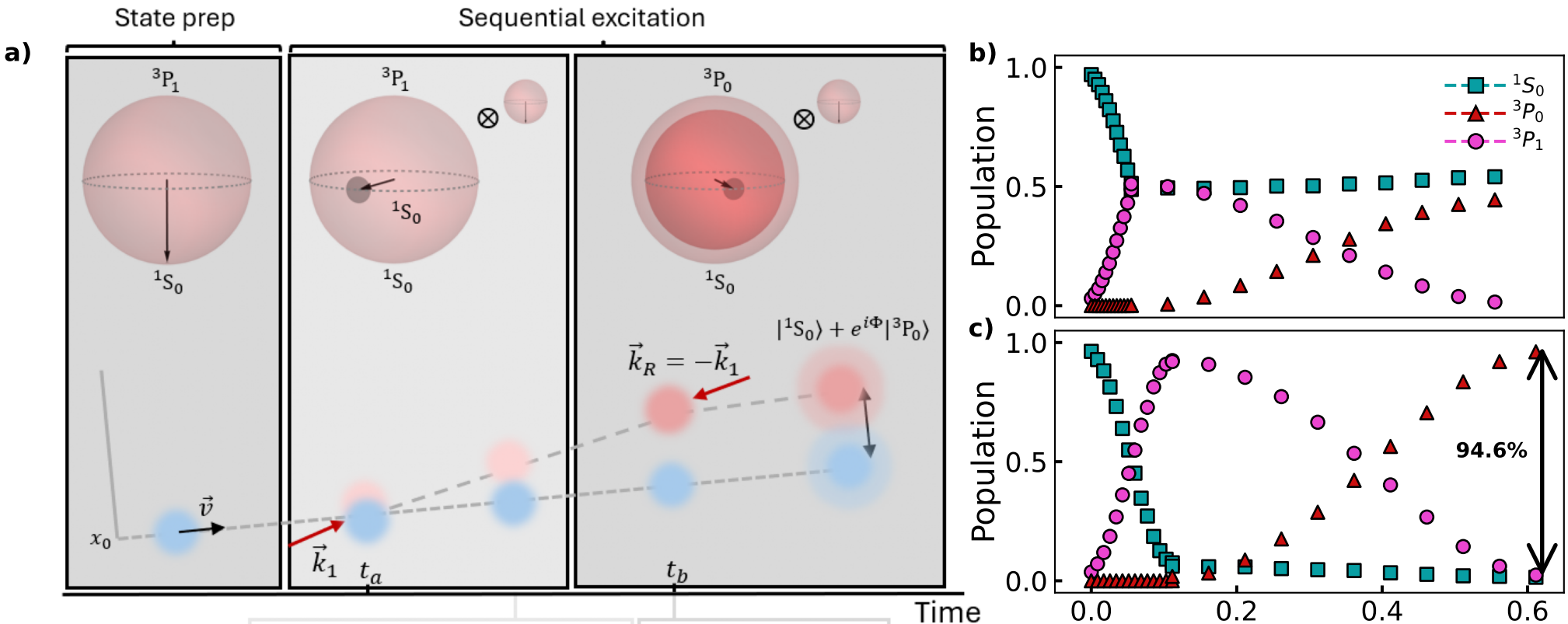

$$\Phi = \vec{k}\cdot(\vec{x}_0 + \vec{v}t_a) + \phi_a - \omega_e(t_b - t_a) - \vec{k}\cdot(\vec{x}_0 + \vec{v}t_b) + \phi_b$$